

NTA JEE Mains Jan 2026

Test Date	23/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : Mathematics Section A

Q.1 If the points of intersection of the ellipses $x^2 + 2y^2 - 6x - 12y + 23 = 0$ and $4x^2 + 2y^2 - 20x - 12y + 35 = 0$ lie on a circle of radius r and centre (a, b) , then the value of $ab + 18r^2$ is

- Options 1. 55
2. 53
3. 51
4. 52

Question Type : MCQ

Question ID : 444792460

Option 1 ID : 4447921568

Option 2 ID : 4447921569

Option 3 ID : 4447921567

Option 4 ID : 4447921570

Q.2 Let PQ be a chord of the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$, perpendicular to the x-axis such that OPQ is an equilateral triangle, O being the centre of the hyperbola. If the eccentricity of the hyperbola is $\sqrt{3}$, then the area of the triangle OPQ is

- Options 1. $\frac{8\sqrt{3}}{5}$
2. $\frac{11}{5}$
3. $2\sqrt{3}$
4. $\frac{9}{5}$

Question Type : MCQ

Question ID : 444792462

Option 1 ID : 4447921575

Option 2 ID : 4447921578

Option 3 ID : 4447921576

Option 4 ID : 4447921577

Q.3 Let $A = \{0, 1, 2, \dots, 9\}$. Let R be a relation on A defined by $(x, y) \in R$ if and only if $|x - y|$ is a multiple of 3.

Given below are two statements:

Statement I: $n(R) = 36$.

Statement II: R is an equivalence relation.

In the light of the above statements, choose the *correct* answer from the options given below

- Options
1. Statement I is correct but Statement II is incorrect
 2. Statement I is incorrect but Statement II is correct
 3. Both Statement I and Statement II are incorrect
 4. Both Statement I and Statement II are correct

Question Type : **MCQ**

Question ID : **444792451**

Option 1 ID : **4447921533**

Option 2 ID : **4447921534**

Option 3 ID : **4447921532**

Option 4 ID : **4447921531**

Q.4 The area of the region enclosed between the circles $x^2 + y^2 = 4$ and $x^2 + (y - 2)^2 = 4$ is:

- Options
1. $\frac{4}{3}(2\pi - 3\sqrt{3})$
 2. $\frac{2}{3}(2\pi - 3\sqrt{3})$
 3. $\frac{2}{3}(4\pi - 3\sqrt{3})$
 4. $\frac{4}{3}(2\pi - \sqrt{3})$

Question Type : **MCQ**

Question ID : **444792470**

Option 1 ID : **4447921607**

Option 2 ID : **4447921609**

Option 3 ID : **4447921608**

Option 4 ID : **4447921610**

Q.5 The system of linear equations

$$x + y + z = 6$$

$$2x + 5y + az = 36$$

$$x + 2y + 3z = b$$

has

- Options
1. unique solution for $a = 8$ and $b = 14$
 2. infinitely many solutions for $a = 8$ and $b = 14$
 3. infinitely many solutions for $a = 8$ and $b = 16$
 4. unique solution for $a = 8$ and $b = 16$

Question Type : MCQ

Question ID : 444792455

Option 1 ID : 4447921547

Option 2 ID : 4447921549

Option 3 ID : 4447921550

Option 4 ID : 4447921548

Q.6 Consider two sets $A = \{x \in \mathbb{Z} : |(x-3) - 3| \leq 1\}$ and

$$B = \left\{ x \in \mathbb{R} - \{1, 2\} : \frac{(x-2)(x-4)}{x-1} \log_e(|x-2|) = 0 \right\}.$$

Then the number of onto functions $f: A \rightarrow B$ is equal to

- Options
1. 79
 2. 81
 3. 62
 4. 32

Question Type : MCQ

Question ID : 444792452

Option 1 ID : 4447921537

Option 2 ID : 4447921538

Option 3 ID : 4447921536

Option 4 ID : 4447921535

Q.7 Bag A contains 9 white and 8 black balls, while bag B contains 6 white and 4 black balls. One ball is randomly picked up from the bag B and mixed up with the balls in the bag A. Then a ball is randomly drawn from the bag A. If the probability, that the ball drawn is white, is $\frac{p}{q}$, $\gcd(p, q) = 1$, then $p + q$ is equal to

- Options
1. 23
 2. 22
 3. 21
 4. 24

Question Type : MCQ

Question ID : 444792458

Option 1 ID : 4447921559

Option 2 ID : 4447921560

Option 3 ID : 4447921561

Option 4 ID : 4447921562

Q.8

Let $I(x) = \int \frac{3dx}{(4x+6)(\sqrt{4x^2+8x+3})}$ and $I(0) = \frac{\sqrt{3}}{4} + 20$. If

$I\left(\frac{1}{2}\right) = \frac{a\sqrt{2}}{b} + c$, where $a, b, c \in \mathbb{N}$, $\gcd(a, b) = 1$, then $a + b + c$ is equal to

Options 1. 28

2. 30

3. 29

4. 31

Question Type : MCQ

Question ID : 444792469

Option 1 ID : 4447921603

Option 2 ID : 4447921605

Option 3 ID : 4447921604

Option 4 ID : 4447921606

Q.9

Let $\frac{\pi}{2} < \theta < \pi$ and $\cot \theta = -\frac{1}{2\sqrt{2}}$. Then the value of

$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta + \sin 8\theta) + \cos\left(\frac{15\theta}{2}\right)(\cos 8\theta - \sin 8\theta)$

is equal to

Options

1. $\frac{\sqrt{2}}{\sqrt{3}}$ 2. $-\frac{\sqrt{2}}{\sqrt{3}}$ 3. $\frac{\sqrt{2}-1}{\sqrt{3}}$ 4. $\frac{1-\sqrt{2}}{\sqrt{3}}$

Question Type : MCQ

Question ID : 444792464

Option 1 ID : 4447921585

Option 2 ID : 4447921586

Option 3 ID : 4447921583

Option 4 ID : 4447921584

Q.10

Let $\sum_{k=1}^n a_k = \alpha n^2 + \beta n$. If $a_{10} = 59$ and $a_6 = 7a_1$, then $\alpha + \beta$ is equal to

Options 1. 7

2. 3

3. 12

4. 5

Question Type : MCQ

Question ID : 444792456

Option 1 ID : 4447921553

Option 2 ID : 4447921551

Option 3 ID : 4447921554

Option 4 ID : 4447921552

Q.11 If the mean and the variance of the data

Class	4-8	8-12	12-16	16-20
Frequency	3	λ	4	7

are μ and 19 respectively, then the value of $\lambda + \mu$ is

- Options 1. 19
2. 21
3. 20
4. 18

Question Type : MCQ

Question ID : 444792459

Option 1 ID : 4447921564

Option 2 ID : 4447921566

Option 3 ID : 4447921565

Option 4 ID : 4447921563

Q.12 The sum of all the real solutions of the equation

$$\log_{(x+3)}(6x^2 + 28x + 30) = 5 - 2 \log_{(6x+10)}(x^2 + 6x + 9) \text{ is equal to}$$

- Options 1. 4
2. 2
3. 1
4. 0

Question Type : MCQ

Question ID : 444792453

Option 1 ID : 4447921542

Option 2 ID : 4447921541

Option 3 ID : 4447921540

Option 4 ID : 4447921539

Q.13 Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{c} = \lambda\hat{i} + \hat{j} + \hat{k}$ and $\vec{v} = \vec{a} \times \vec{b}$. If

$\vec{v} \cdot \vec{c} = 11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p , then $9p^2$ is equal to

- Options 1. 6
2. 12
3. 4
4. 9

Question Type : MCQ

Question ID : 444792465

Option 1 ID : 4447921588

Option 2 ID : 4447921590

Option 3 ID : 4447921587

Option 4 ID : 4447921589

Q.14 An equilateral triangle OAB is inscribed in the parabola $y^2 = 4x$ with the vertex O at the vertex of the parabola. Then the minimum distance of the circle having AB as a diameter from the origin is

- Options
1. $2(3 + \sqrt{3})$
 2. $4(3 - \sqrt{3})$
 3. $2(8 - 3\sqrt{3})$
 4. $4(6 + \sqrt{3})$

Question Type : MCQ

Question ID : 444792461

Option 1 ID : 4447921573

Option 2 ID : 4447921574

Option 3 ID : 4447921571

Option 4 ID : 4447921572

Q.15 The least value of $(\cos^2 \theta - 6\sin \theta \cos \theta + 3\sin^2 \theta + 2)$ is

- Options
1. -1
 2. 1
 3. $4 - \sqrt{10}$
 4. $4 + \sqrt{10}$

Question Type : MCQ

Question ID : 444792468

Option 1 ID : 4447921601

Option 2 ID : 4447921602

Option 3 ID : 4447921600

Option 4 ID : 4447921599

Q.16 If $f(x) = \begin{cases} \frac{a|x| + x^2 - 2(\sin|x|)(\cos|x|)}{x} & , x \neq 0 \\ b & , x = 0 \end{cases}$

is continuous at $x = 0$, then $a + b$ is equal to

- Options
1. 4
 2. 0
 3. 2
 4. 1

Question Type : MCQ

Question ID : 444792467

Option 1 ID : 4447921598

Option 2 ID : 4447921595

Option 3 ID : 4447921597

Option 4 ID : 4447921596

Q.17 Let A(1, 2) and C(-3, -6) be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x - y = 14$. If B (α , β) and D (γ , δ) are the other two vertices, then $|\alpha + \beta + \gamma + \delta|$ is equal to

- Options 1. 6
2. 3
3. 9
4. 1

Question Type : MCQ

Question ID : 444792463

Option 1 ID : 4447921581

Option 2 ID : 4447921580

Option 3 ID : 4447921582

Option 4 ID : 4447921579

Q.18 If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}$, $i = \sqrt{-1}$, then $(z^{201} - i)^8$ is equal to

- Options 1. 1
2. 0
3. -1
4. 256

Question Type : MCQ

Question ID : 444792454

Option 1 ID : 4447921545

Option 2 ID : 4447921543

Option 3 ID : 4447921544

Option 4 ID : 4447921546

Q.19 Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three vectors such that $\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{c})$. If $|\vec{a}| = 1$, $|\vec{b}| = 4$, $|\vec{c}| = 2$, and the angle between $\vec{b}$ and $\vec{c}$ is 60° , then $|\vec{a} \cdot \vec{c}|$ is equal to

- Options 1. 0
2. 4
3. 1
4. 2

Question Type : MCQ

Question ID : 444792466

Option 1 ID : 4447921591

Option 2 ID : 4447921594

Option 3 ID : 4447921592

Option 4 ID : 4447921593

Q.20 The number of ways, in which 16 oranges can be distributed to four children such that each child gets at least one orange, is

- Options 1. 384
2. 429
3. 455
4. 403

Question Type : MCQ

Question ID : 444792457

Option 1 ID : 4447921555

Option 2 ID : 4447921558

Option 3 ID : 4447921557

Option 4 ID : 4447921556

Section : Mathematics Section B

Q.21 If the image of the point $P(a, 2, a)$ in the line $\frac{x}{2} = \frac{y+a}{1} = \frac{z}{1}$ is Q and the image of Q in the line $\frac{x-2b}{2} = \frac{y-a}{1} = \frac{z+2b}{-5}$ is P, then $a + b$ is equal to ____.

Question Type : SA

Question ID : 444792473

Q.22 Let $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$ and B be a matrix such that $B(I - A) = I + A$. Then the sum of the diagonal elements of $B^T B$ is equal to ____

Question Type : SA

Question ID : 444792471

Q.23 The number of elements in the

$$\text{set } S = \left\{ x : x \in [0, 100] \text{ and } \int_0^x t^2 \sin(x-t) dt = x^2 \right\} \text{ is } \underline{\hspace{2cm}}$$

Question Type : SA

Question ID : 444792474

Q.24 Let S denote the set of 4-digit numbers $abcd$ such that $a > b > c > d$ and P denote the set of 5-digit numbers having product of its digits equal to 20. Then $n(S) + n(P)$ is equal to ____

Question Type : SA

Question ID : 444792472

Q.25 If the solution curve $y = f(x)$ of the differential equation $(x^2 - 4)y' - 2xy + 2x(4 - x^2)^2 = 0$, $x > 2$, passes through the point $(3, 15)$, then the local maximum value of f is ____

Question Type : SA

Question ID : 444792475

Section : Physics Section A

Q.26 To compare EMF of two cells using potentiometer the balancing lengths obtained are 200 cm and 150 cm. The least count of scale is 1 cm. The percentage error in the ratio of EMFs is _____

- Options
1. 1.65
 2. 1.45
 3. 1.75
 4. 1.55

Question Type : **MCQ**

Question ID : **444792476**

Option 1 ID : **4447921618**

Option 2 ID : **4447921616**

Option 3 ID : **4447921619**

Option 4 ID : **4447921617**

Q.27 An air bubble of volume 2.9 cm^3 rises from the bottom of a swimming pool of 5 m deep. At the bottom of the pool water temperature is 17°C . The volume of the bubble when it reaches the surface, where the water temperature is 27°C , is _____ cm^3 .

($g = 10 \text{ m/s}^2$, density of water = 10^3 kg/m^3 , and 1 atm pressure is 10^5 Pa)

- Options
1. 4.2
 2. 2.0
 3. 4.5
 4. 3.0

Question Type : **MCQ**

Question ID : **444792482**

Option 1 ID : **4447921640**

Option 2 ID : **4447921641**

Option 3 ID : **4447921643**

Option 4 ID : **4447921642**

Q.28 A block is sliding down on an inclined plane of slope θ and at an instant $t = 0$ this block is given an upward momentum so that it starts moving up on the inclined surface with velocity u . The distance (S) travelled by the block before its velocity become zero, is _____.
(g = gravitational acceleration)

Options

1. $\frac{u^2}{4g \sin \theta}$
2. $\frac{u^2}{2g \cos \theta}$
3. $\frac{u^2}{\sqrt{2}g \cos \theta}$
4. $\frac{2u^2}{g \cos \theta}$

Question Type : **MCQ**

Question ID : **444792479**

Option 1 ID : **4447921628**

Option 2 ID : **4447921629**

Option 3 ID : **4447921631**

Option 4 ID : **4447921630**

Q.29 When an unpolarized light falls at a particular angle on a glass plate (placed in air), it is observed that the reflected beam is linearly polarized. The angle of refracted beam with respect to the normal is _____.

($\tan^{-1}(1.52) = 57.7^\circ$, refractive indices of air and glass are 1.00 and 1.52, respectively.)

Options

1. 32.3°
2. 42.6°
3. 39.6°
4. 36.3°

Question Type : **MCQ**

Question ID : **444792492**

Option 1 ID : **4447921680**

Option 2 ID : **4447921683**

Option 3 ID : **4447921682**

Option 4 ID : **4447921681**

Q.30 A prism of angle 75° and refractive index $\sqrt{3}$ is coated with thin film of refractive index 1.5 only at the back exit surface. To have total internal reflection at the back exit surface the incident angle must be _____.
($\sin 15^\circ = 0.25$ and $\sin 25^\circ = 0.43$)

- Options 1. between 15° and 20°
2. $< 15^\circ$
3. 15°
4. $> 25^\circ$

Question Type : MCQ

Question ID : 444792493

Option 1 ID : 4447921686

Option 2 ID : 4447921687

Option 3 ID : 4447921684

Option 4 ID : 4447921685

Q.31 A circular loop of radius 7 cm is placed in uniform magnetic field of 0.2 T directed perpendicular to plane of loop. The loop is converted into a square loop in 0.5 s. The EMF induced in the loop is _____ mV.

- Options 1. 6.6
2. 8.25
3. 13.2
4. 1.32

Question Type : MCQ

Question ID : 444792490

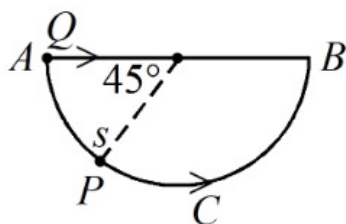
Option 1 ID : 4447921674

Option 2 ID : 4447921675

Option 3 ID : 4447921673

Option 4 ID : 4447921672

Q.32 A bead P sliding on a frictionless semi-circular string (ACB) and it is at point S at $t = 0$ and at this instant the horizontal component of its velocity is v . Another bead Q of the same mass as P is ejected from point A at $t = 0$ along the horizontal string AB , with the speed v , friction between the beads and the respective strings may be neglected in both cases. Let t_P and t_Q be the respective times taken by beads P and Q to reach the point B , then the relation between t_P and t_Q is



- Options 1. $t_P = t_Q$
2. $t_P < t_Q$
3. $t_P > t_Q$
4. $t_P > 1.25t_Q$

Question Type : MCQ

Question ID : 444792478

Option 1 ID : 4447921626

Option 2 ID : 4447921624

Option 3 ID : 4447921625

Option 4 ID : 4447921627

Q.33 Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10 \sin(\omega t)$ A, where $\omega = 1000$ rad./s. A circular conducting loop (B) of radius 1 cm coaxially slid through the solenoid at a speed $v = 1$ cm/s. The r.m.s. current through the loop when the coil B is inserted 10 cm inside the solenoid is $\alpha / \sqrt{2}$ μ A. The value of α is _____.
[Resistance of the loop = 10 Ω]

Options 1. 197

2. 100

3. 80

4. 280

Question Type : MCQ

Question ID : 444792486

Option 1 ID : 4447921658

Option 2 ID : 4447921656

Option 3 ID : 4447921659

Option 4 ID : 4447921657

Q.34 The internal energy of a monoatomic gas is $3nRT$. One mole of helium is kept in a cylinder having internal cross section area of 17 cm² and fitted with a light movable frictionless piston. The gas is heated slowly by supplying 126 J heat. If the temperature rises by 4 °C, then the piston will move _____ cm.
(atmospheric pressure = 10⁵ Pa)

Options 1. 1.45

2. 1.55

3. 14.5

4. 15.5

Question Type : MCQ

Question ID : 444792484

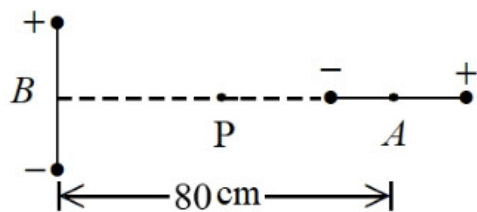
Option 1 ID : 4447921651

Option 2 ID : 4447921650

Option 3 ID : 4447921649

Option 4 ID : 4447921648

Q.35 Two short dipoles (A, B), A having charges $\pm 2 \mu\text{C}$ and length 1 cm and B having charges $\pm 4 \mu\text{C}$ and length 1 cm are placed with their centres 80 cm apart as shown in the figure. The electric field at a point P , equi-distant from the centres of both dipoles is _____ N/C .



Options

1. $9\sqrt{2} \times 10^4$
2. $4.5\sqrt{2} \times 10^4$
3. $\frac{9}{16}\sqrt{2} \times 10^4$
4. $\frac{9}{16}\sqrt{2} \times 10^5$

Question Type : MCQ

Question ID : 444792488

Option 1 ID : 4447921664

Option 2 ID : 4447921667

Option 3 ID : 4447921665

Option 4 ID : 4447921666

Q.36 The ratio of speeds of electromagnetic waves in vacuum and a medium, having dielectric constant $k = 3$ and permeability of $\mu = 2\mu_0$, is (μ_0 = permeability of vacuum)

Options

1. $36 : 1$
2. $6 : 1$
3. $3 : 2$
4. $\sqrt{6} : 1$

Question Type : MCQ

Question ID : 444792491

Option 1 ID : 4447921678

Option 2 ID : 4447921677

Option 3 ID : 4447921676

Option 4 ID : 4447921679

Q.37 The current passing through a conducting loop in the form of equilateral triangle of side $4\sqrt{3}$ cm is 2 A. The magnetic field at its centroid is $\alpha \times 10^{-5}$ T. The value of α is _____.
(Given : $\mu_0 = 4\pi \times 10^{-7}$ SI units)

Options

1. $\frac{\sqrt{3}}{2}$
2. $\sqrt{3}$
3. $3\sqrt{3}$
4. $2\sqrt{3}$

Question Type : MCQ

Question ID : 444792487

Option 1 ID : 4447921663

Option 2 ID : 4447921660

Option 3 ID : 4447921662

Option 4 ID : 4447921661

Q.38 A body of mass 14 kg initially at rest explodes and breaks into three fragments of masses in the ratio 2 : 2 : 3. The two pieces of equal masses fly off perpendicular to each other with a speed of 18 m/s each. The velocity of the heavier fragment is _____ m/s.

Options

1. 12
2. $10\sqrt{2}$
3. $12\sqrt{2}$
4. $24\sqrt{2}$

Question Type : MCQ

Question ID : 444792480

Option 1 ID : 4447921634

Option 2 ID : 4447921633

Option 3 ID : 4447921632

Option 4 ID : 4447921635

Q.39 Which of the following pair of nuclei are isobars of the element?

Options

1. $^{198}_{80}\text{Hg}$ and $^{197}_{79}\text{Au}$
2. ^2_1H and ^3_1H
3. ^3_1H and ^3_2He
4. $^{236}_{92}\text{U}$ and $^{238}_{92}\text{U}$

Question Type : MCQ

Question ID : 444792494

Option 1 ID : 4447921691

Option 2 ID : 4447921688

Option 3 ID : 4447921690

Option 4 ID : 4447921689

Q.40 A parallel plate capacitor with plate separation 5 mm is charged by a battery. On introducing a mica sheet of 2 mm and maintaining the connections of the plates with the terminals of the battery, it is found that it draws 25% more charge from the battery. The dielectric constant of mica is _____.

- Options 1. 1.0
2. 1.5
3. 2.0
4. 2.5

Question Type : MCQ

Question ID : 444792485

Option 1 ID : 4447921653

Option 2 ID : 4447921654

Option 3 ID : 4447921655

Option 4 ID : 4447921652

Q.41 One mole of an ideal diatomic gas expands from volume V to $2V$ isothermally at a temperature 27°C and does W joule of work. If the gas undergoes same magnitude of expansion adiabatically from 27°C doing the same amount of work W , then its final temperature will be (close to) _____ $^\circ\text{C}$.
($\log_e 2 = 0.693$)

- Options 1. -30
2. -56
3. -117
4. -189

Question Type : MCQ

Question ID : 444792483

Option 1 ID : 4447921646

Option 2 ID : 4447921645

Option 3 ID : 4447921644

Option 4 ID : 4447921647

Q.42 A small metallic sphere of diameter 2 mm and density 10.5 g/cm^3 is dropped in glycerine having viscosity 10 Poise and density 1.5 g/cm^3 respectively. The terminal velocity attained by the sphere is _____ cm/s.
($\pi = \frac{22}{7}$ and $g = 10\text{ m/s}^2$)

- Options 1. 3.0
2. 1.5
3. 1.0
4. 2.0

Question Type : MCQ

Question ID : 444792481

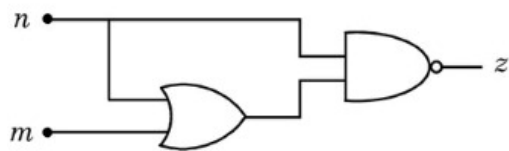
Option 1 ID : 4447921639

Option 2 ID : 4447921637

Option 3 ID : 4447921636

Option 4 ID : 4447921638

Q.43 For the given logic gate circuit, which of the following is the correct truth table ?



Options

<i>n</i>	<i>m</i>	<i>z</i>
0	0	1
0	1	0
1	1	1
1	0	0

<i>n</i>	<i>m</i>	<i>z</i>
0	0	0
0	1	1
1	1	0
1	0	1

<i>n</i>	<i>m</i>	<i>z</i>
0	0	1
0	1	0
1	1	0
1	0	0

<i>n</i>	<i>m</i>	<i>z</i>
0	0	1
0	1	1
1	1	0
1	0	0

Question Type : MCQ

Question ID : 444792495

Option 1 ID : 4447921693

Option 2 ID : 4447921695

Option 3 ID : 4447921692

Option 4 ID : 4447921694

Q.44 A paratrooper jumps from an aeroplane and opens a parachute after 2 s of free fall and starts deaccelerating with 3 m/s^2 . At 10 m height from ground, while descending with the help of parachute, the speed of paratrooper is 5 m/s. The initial height of the airplane is _____ m.
($g = 10 \text{ m/s}^2$)

- Options 1. 20
2. 92.5
3. 62.5
4. 82.5

Question Type : **MCQ**

Question ID : **444792477**

Option 1 ID : **4447921623**

Option 2 ID : **4447921622**

Option 3 ID : **4447921621**

Option 4 ID : **4447921620**

Q.45 Two charges $7 \mu\text{C}$ and $-2 \mu\text{C}$ are placed at $(-9, 0, 0) \text{ cm}$ and $(9, 0, 0) \text{ cm}$ respectively in an external field $E = \frac{A}{r^2} \hat{r}$, where $A = 9 \times 10^5 \text{ N/C.m}^2$.
Considering the potential at infinity is 0, the electrostatic energy of the configuration is _____ J.

- Options 1. 1.4
2. 24.3
3. 49.3
4. -90.7

Question Type : **MCQ**

Question ID : **444792489**

Option 1 ID : **4447921669**

Option 2 ID : **4447921671**

Option 3 ID : **4447921670**

Option 4 ID : **4447921668**

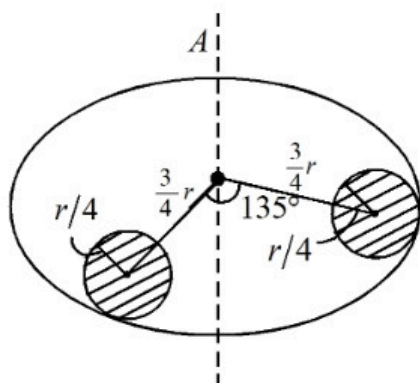
Section : **Physics Section B**

Q.46 The velocity of sound in air is doubled when the temperature is raised from 0°C to $\alpha^\circ\text{C}$. The value of α is _____.

Question Type : **SA**

Question ID : **444792498**

Q.47 Suppose there is a uniform circular disc of mass M kg and radius r m shown in figure. The shaded regions are cut out from the disc. The moment of inertia of the remainder about the axis A of the disc is given by $\frac{x}{256} Mr^2$. The value of x is _____.



Question Type : SA

Question ID : 444792496

Q.48 A ball of radius r and density ρ dropped through a viscous liquid of density σ and viscosity η attains its terminal velocity at time t , given by $t = A \rho^a r^b \eta^c \sigma^d$, where A is a constant and a, b, c and d are integers. The value of $\frac{b+c}{a+d}$ is _____.

Question Type : SA

Question ID : 444792497

Q.49 The average energy released per fission for the nucleus of ${}^{235}_{92}\text{U}$ is 190 MeV. When all the atoms of 47 g pure ${}^{235}_{92}\text{U}$ undergo fission process, the energy released is $\alpha \times 10^{23}$ MeV. The value of α is _____.

(Avogadro Number = 6×10^{23} per mole)

Question Type : SA

Question ID : 444792500

Q.50 The size of the images of an object, formed by a thin lens are equal when the object is placed at two different positions 8 cm and 24 cm from the lens. The focal length of the lens is _____ cm.

Question Type : SA

Question ID : 444792499

Section : Chemistry Section A

Q.51 Identify the **CORRECT** set of details from the following:

- A. $[\text{Co}(\text{NH}_3)_6]^{3+}$: Inner orbital complex; d^2sp^3 hybridized
- B. $[\text{MnCl}_6]^{3-}$: Outer orbital complex; sp^3d^2 hybridized
- C. $[\text{CoF}_6]^{3-}$: Outer orbital complex; d^2sp^3 hybridized
- D. $[\text{FeF}_6]^{3-}$: Outer orbital complex; sp^3d^2 hybridized
- E. $[\text{Ni}(\text{CN})_4]^{2-}$: Inner orbital complex; sp^3 hybridized

Choose the correct answer from the options given below:

- Options
- 1. A, B & D Only
 - 2. C & D Only
 - 3. A, B, C, D & E
 - 4. A, C & E Only

Question Type : MCQ

Question ID : 444792511

Option 1 ID : 4447921742

Option 2 ID : 4447921744

Option 3 ID : 4447921741

Option 4 ID : 4447921743

Q.52 Given below are two statements:

Statement I: The second ionisation enthalpy of Na is larger than the corresponding ionisation enthalpy of Mg.

Statement II: The ionic radius of O^{2-} is larger than that of F^- .

In the light of the above statements, choose the *correct* answer from the options given below

- Options
- 1. Statement I is false but Statement II is true
 - 2. Both Statement I and Statement II are true
 - 3. Statement I is true but Statement II is false
 - 4. Both Statement I and Statement II are false

Question Type : MCQ

Question ID : 444792508

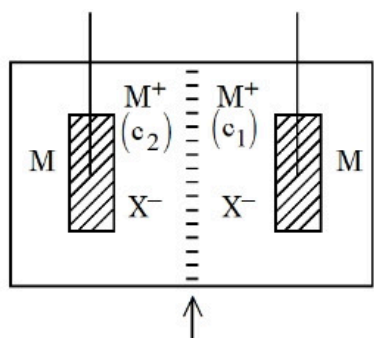
Option 1 ID : 4447921732

Option 2 ID : 4447921729

Option 3 ID : 4447921731

Option 4 ID : 4447921730

Q.53



Semi permeable membrane

Consider the above electrochemical cell where a metal electrode (M) is undergoing redox reaction by forming M^+ ($M \rightarrow M^+ + e^-$). The cation M^+ is present in two different concentrations c_1 and c_2 as shown above. Which of the following statement is correct for generating a positive cell potential?

- Options
1. If c_1 is present at anode, then $c_1 = c_2$.
 2. If c_1 is present at cathode, then $c_1 > c_2$.
 3. If c_1 is present at anode, then $c_1 > c_2$.
 4. If c_1 is present at cathode, then $c_1 < c_2$.

Question Type : MCQ

Question ID : 444792505

Option 1 ID : 4447921720

Option 2 ID : 4447921719

Option 3 ID : 4447921718

Option 4 ID : 4447921717

Q.54 Identify the INCORRECT statements from the following:

- A. Notation ${}^{24}_{12}\text{Mg}$ represents 24 protons and 12 neutrons.
- B. Wavelength of a radiation of frequency $4.5 \times 10^{15} \text{ s}^{-1}$ is $6.7 \times 10^{-8} \text{ m}$.
- C. One radiation has wavelength = λ_1 (900 nm) and energy = E_1 . Other radiation has wavelength = λ_2 (300 nm) and energy = E_2 . $E_1 : E_2 = 3 : 1$.
- D. Number of photons of light of wavelength 2000 pm that provides 1 J of energy is 1.006×10^{16} .

Choose the **correct** answer from the options given below:

- Options
1. A and B Only
 2. A and C Only
 3. A and D Only
 4. B and C Only

Question Type : MCQ

Question ID : 444792502

Option 1 ID : 4447921705

Option 2 ID : 4447921706

Option 3 ID : 4447921707

Option 4 ID : 4447921708

Q.55 A student has been given a compound "x" of molecular formula- C_6H_7N . 'x' is sparingly soluble in water. However, on addition of dilute mineral acid, 'x' becomes soluble in water. 'x' when treated with $CHCl_3$ and $KOH(alc)$, 'y' is produced. 'y' has a specific unpleasant smell. On treatment with benzenesulphonyl chloride, 'x' gives a compound 'z' which is soluble in alkali. The number of different "H" atoms present in 'z' is:-

- Options 1. 5
2. 8
3. 4
4. 7

Question Type : **MCQ**

Question ID : **444792518**

Option 1 ID : **4447921769**

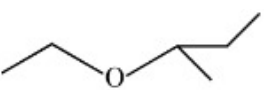
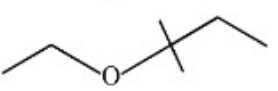
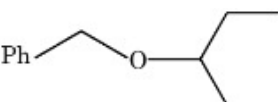
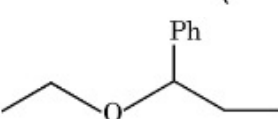
Option 2 ID : **4447921772**

Option 3 ID : **4447921770**

Option 4 ID : **4447921771**

Q.56 A mixed ether (P), when heated with excess of hot concentrated hydrogen iodide produces two different alkyl iodides which when treated with aq. NaOH give compounds (Q) and (R). Both (Q) and (R) give yellow precipitate with NaOI. Identify the mixed ether (P):

Options

1. 
2. 
3. 
4. 

Question Type : **MCQ**

Question ID : **444792516**

Option 1 ID : **4447921763**

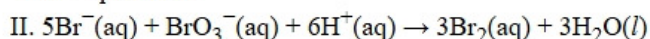
Option 2 ID : **4447921761**

Option 3 ID : **4447921762**

Option 4 ID : **4447921764**

Q.57 Observe the following reactions at T(K).

I. $A \rightarrow \text{products}$.



Both the reactions are started at 10.00 am. The rates of these reactions at 10.10 am are same. The value of $-\frac{\Delta[\text{Br}^-]}{\Delta t}$ at 10.10 am is $2 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}$. The concentration of A at 10.10 am is $10^{-2} \text{ mol L}^{-1}$. What is the first order rate constant (in min^{-1}) of reaction I?

- Options
1. 2×10^{-3}
 2. 10^{-3}
 3. 10^{-2}
 4. 4×10^{-3}

Question Type : MCQ

Question ID : 444792507

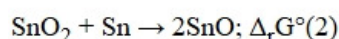
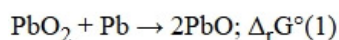
Option 1 ID : 4447921726

Option 2 ID : 4447921728

Option 3 ID : 4447921727

Option 4 ID : 4447921725

Q.58 It is noticed that Pb^{2+} is more stable than Pb^{4+} but Sn^{2+} is less stable than Sn^{4+} . Observe the following reactions.



Identify the correct set from the following

- Options
1. $\Delta_r G^\circ(1) < 0$; $\Delta_r G^\circ(2) < 0$
 2. $\Delta_r G^\circ(1) > 0$; $\Delta_r G^\circ(2) > 0$
 3. $\Delta_r G^\circ(1) < 0$; $\Delta_r G^\circ(2) > 0$
 4. $\Delta_r G^\circ(1) > 0$; $\Delta_r G^\circ(2) < 0$

Question Type : MCQ

Question ID : 444792504

Option 1 ID : 4447921714

Option 2 ID : 4447921715

Option 3 ID : 4447921713

Option 4 ID : 4447921716

Q.59 The oxidation state of chromium in the final product formed in the reaction between KI and acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution is:

- Options
1. +4
 2. +2
 3. +3
 4. +6

Question Type : MCQ

Question ID : 444792510

Option 1 ID : 4447921737

Option 2 ID : 4447921739

Option 3 ID : 4447921740

Option 4 ID : 4447921738

Q.60 Which statements are **NOT TRUE** about XeO_2F_2 ?

- A. It has a see-saw shape.
- B. Xe has 5 electron pairs in its valence shell in XeO_2F_2 .
- C. The O-Xe-O bond angle is close to 180° .
- D. The F-Xe-F bond angle is close to 180° .
- E. Xe has 16 valence electrons in XeO_2F_2 .

Choose the **correct** answer from the options given below:

Options 1. B and D Only

- 2. A and D Only
- 3. B, D and E Only
- 4. B, C and E Only

Question Type : MCQ

Question ID : 444792503

Option 1 ID : 4447921710

Option 2 ID : 4447921711

Option 3 ID : 4447921712

Option 4 ID : 4447921709

Q.61 Given below are two statements:

Statement I: $(\text{CH}_3)_3\text{C}^\oplus$ is more stable than $\text{CH}_3^\oplus$ as nine hyperconjugation interactions are possible in $(\text{CH}_3)_3\text{C}^\oplus$.

Statement II: $\text{CH}_3^\oplus$ is less stable than $(\text{CH}_3)_3\text{C}^\oplus$ as only three hyperconjugation interactions are possible in $\text{CH}_3^\oplus$.

In the light of the above statements, choose the **correct** answer from the options given below

- Options 1. Statement I is true but Statement II is false
- 2. Both Statement I and Statement II are true
 - 3. Statement I is false but Statement II is true
 - 4. Both Statement I and Statement II are false

Question Type : MCQ

Question ID : 444792513

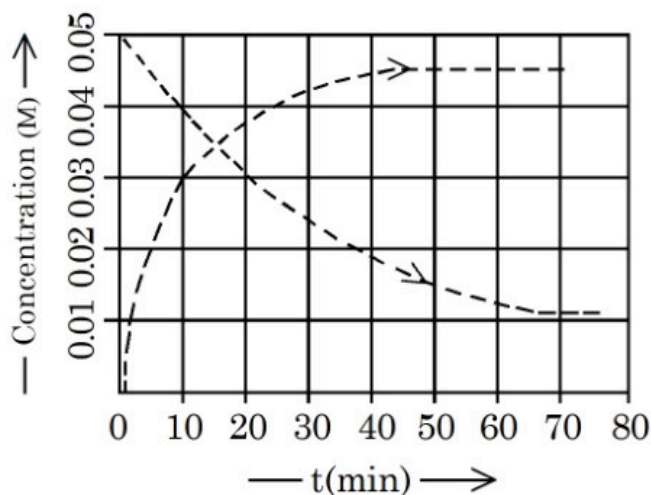
Option 1 ID : 4447921751

Option 2 ID : 4447921749

Option 3 ID : 4447921752

Option 4 ID : 4447921750

Q.62



Given above is the concentration vs time plot for a dissociation reaction : $A \rightarrow nB$.

Based on the data of the initial phase of the reaction (initial 10 min), the value of n is _____.

Options 1. 3

2. 2

3. 4

4. 5

Question Type : MCQ

Question ID : 444792506

Option 1 ID : 4447921723

Option 2 ID : 4447921724

Option 3 ID : 4447921721

Option 4 ID : 4447921722

Q.63 Iodoform test can differentiate between

- A. Methanol and Ethanol
- B. CH_3COOH and $\text{CH}_3\text{CH}_2\text{COOH}$
- C. Cyclohexene and cyclohexanone
- D. Diethyl ether and Pentan-3-one
- E. Anisole and acetone

Choose the **correct** answer from the options given below:

Options 1. A & D Only

2. A & E Only

3. A, B & E Only

4. B, C & E Only

Question Type : MCQ

Question ID : 444792520

Option 1 ID : 4447921780

Option 2 ID : 4447921779

Option 3 ID : 4447921777

Option 4 ID : 4447921778

Q.64 Elements X and Y belong to Group 15. The difference between the electronegativity values of 'X' and phosphorus is higher than that of the difference between phosphorus and 'Y'. 'X' & 'Y' are respectively

- Options
1. As & Bi
 2. Bi & N
 3. As & Sb
 4. N & As

Question Type : **MCQ**

Question ID : **444792509**

Option 1 ID : **4447921734**

Option 2 ID : **4447921733**

Option 3 ID : **4447921736**

Option 4 ID : **4447921735**

Q.65 Both human DNA and RNA are chiral molecules. The chirality in DNA and RNA arises due to the presence of

- Options
1. Chiral phosphate unit
 2. D-sugar component
 3. Base unit
 4. L-sugar component

Question Type : **MCQ**

Question ID : **444792519**

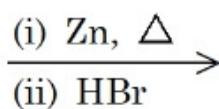
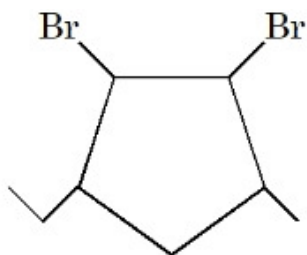
Option 1 ID : **4447921776**

Option 2 ID : **4447921773**

Option 3 ID : **4447921775**

Option 4 ID : **4447921774**

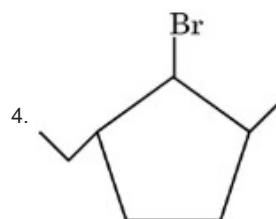
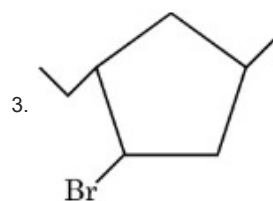
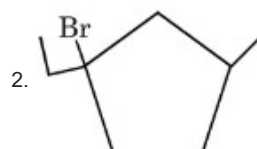
Q.66



(P)
Major Product

Identify (P)

Options



Question Type : MCQ

Question ID : 444792514

Option 1 ID : 4447921754

Option 2 ID : 4447921753

Option 3 ID : 4447921755

Option 4 ID : 4447921756

Q.67 In Carius method 0.2425 g of an organic compound gave 0.5253 g silver chloride.
The percentage of chlorine in the organic compound is

Options 1. 34.79%

2. 37.57%

3. 87.65%

4. 53.58%

Question Type : MCQ

Question ID : 444792512

Option 1 ID : 4447921746

Option 2 ID : 4447921745

Option 3 ID : 4447921748

Option 4 ID : 4447921747

Q.68 The work functions of two metals (M_A and M_B) are in the 1 : 2 ratio. When these metals are exposed to photons of energy 6 eV, the kinetic energy of liberated electrons of $M_A : M_B$ is in the ratio of 2.642 : 1. The work functions (in eV) of M_A and M_B are respectively.

- Options 1. 1.4, 2.8
 2. 3.1, 6.2
 3. 1.5, 3.0
 4. 2.3, 4.6

Question Type : MCQ

Question ID : 444792501

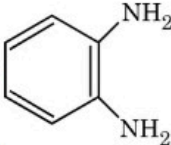
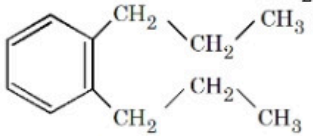
Option 1 ID : 4447921704

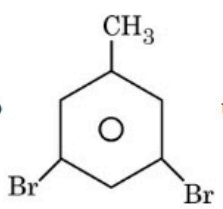
Option 2 ID : 4447921702

Option 3 ID : 4447921703

Option 4 ID : 4447921701

Q.69 Given below are two statements:

Statement I:  can be synthesized from  using simpler reagents in the order i) Acidic $KMnO_4$, ii) Ammonia, iii) Bromine and alkali

Statement II:  can be converted into  using reagents in the order i) Bromine- H_2O ii) $NaNO_2/HCl$ (0 - 5 °C) (iii) Aq. H_3PO_2 .

In the light of the above statements, choose the *correct* answer from the options given below

- Options 1. Statement I is true but Statement II is false
 2. Both Statement I and Statement II are true
 3. Statement I is false but Statement II is true
 4. Both Statement I and Statement II are false

Question Type : MCQ

Question ID : 444792517

Option 1 ID : 4447921767

Option 2 ID : 4447921765

Option 3 ID : 4447921768

Option 4 ID : 4447921766

Q.70 Which of the following statements are **TRUE** about Haloform reaction?:

- A. Sodium hypochlorite reacts with KI to give KOI.
B. KOI is a reducing agent.

C. α, β -unsaturated methylketone ($\text{CH}_3 - \text{CH} = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$) will give

iodoform reaction.

- D. Isopropyl alcohol will not give iodoform test.
E. Methanoic acid will give positive iodoform test.

Choose the **correct** answer from the options given below:

Options 1. B, D & E Only

2. A, B & C Only

3. A, C & E Only

4. A & C Only

Question Type : **MCQ**

Question ID : **444792515**

Option 1 ID : **4447921759**

Option 2 ID : **4447921757**

Option 3 ID : **4447921758**

Option 4 ID : **4447921760**

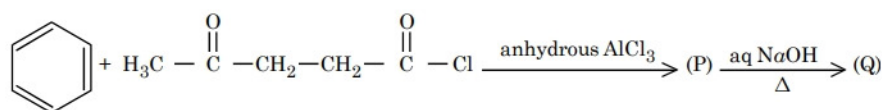
Section : **Chemistry Section B**

Q.71 Two liquids A and B form an ideal solution. At 320 K, the vapour pressure of the solution, containing 3 mol of A and 1 mol of B is 500 mm Hg. At the same temperature, if 1 mol of A is further added to this solution, vapour pressure of the solution increases by 20 mm Hg. Vapour pressure (in mm Hg) of B in pure state is _____. (Nearest integer)

Question Type : **SA**

Question ID : **444792525**

Q.72 Consider the following reaction of benzene.



In compound (Q), the percentage of oxygen is _____. (Nearest integer)

Question Type : **SA**

Question ID : **444792522**

Q.73 200 cc of $x \times 10^{-3}$ M potassium dichromate is required to oxidise 750 cc of 0.6 M Mohr's salt solution in acidic medium.

Here $x =$ _____.

Question Type : **SA**

Question ID : **444792523**

Q.74 Total number of unpaired electrons present in the central metal atoms/ions of $[\text{Ni}(\text{CO})_4]$, $[\text{NiCl}_4]^{2-}$, $[\text{PtCl}_2(\text{NH}_3)_2]$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Pt}(\text{CN})_4]^{2-}$ is _____ .

Question Type : SA

Question ID : 444792521

Q.75 $\text{X}_2(\text{g}) + \text{Y}_2(\text{g}) \rightleftharpoons 2\text{Z}(\text{g})$

$\text{X}_2(\text{g})$ and $\text{Y}_2(\text{g})$ are added to a 1 L flask and it is found that the system attains the above equilibrium at T(K) with the number of moles of $\text{X}_2(\text{g})$, $\text{Y}_2(\text{g})$ and $\text{Z}(\text{g})$ being 3, 3 and 9 mol respectively (equilibrium moles). Under this condition of equilibrium, 10 mol of $\text{Z}(\text{g})$ is added to the flask and the temperature is maintained at T(K). Then the number of moles of $\text{Z}(\text{g})$ in the flask when the new equilibrium is established is _____. (Nearest integer)

Question Type : SA

Question ID : 444792524